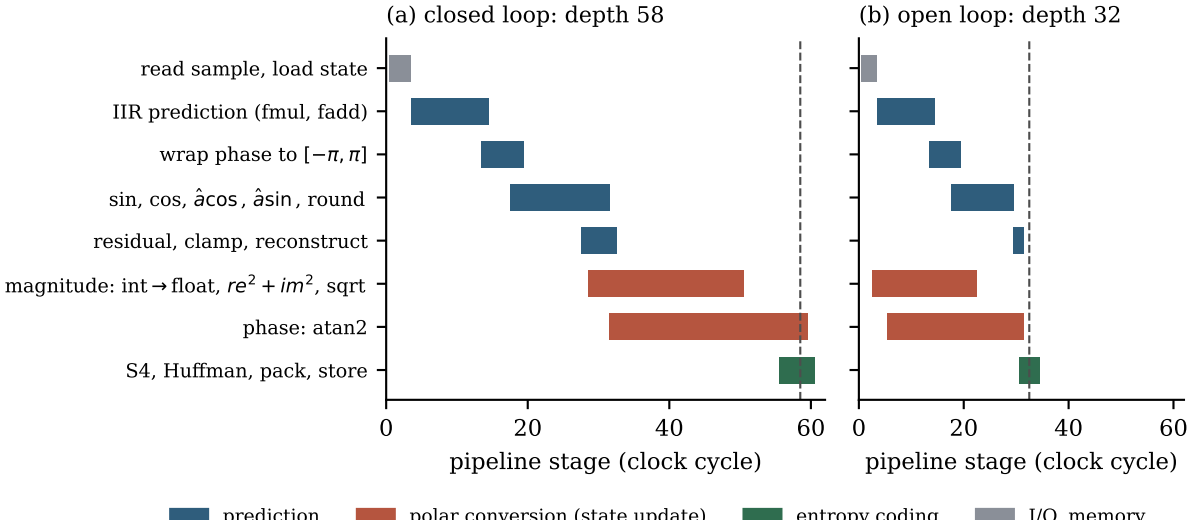
(a) closed loop: depth 58
(b) open loop: depth 32
read sample, load state
IIR prediction (fmul, fadd)
wrap phase to $[-\pi, \pi]$
sin, cos, $\hat{a}$cos, $\hat{a}$sin, round
residual, clamp, reconstruct
magnitude: int → float, $re^2 + im^2$, sqrt
phase: atan2
S4, Huffman, pack, store
0
20
40
60
pipeline stage (clock cycle)
prediction
polar conversion (state update)
entropy coding
I/O, memory